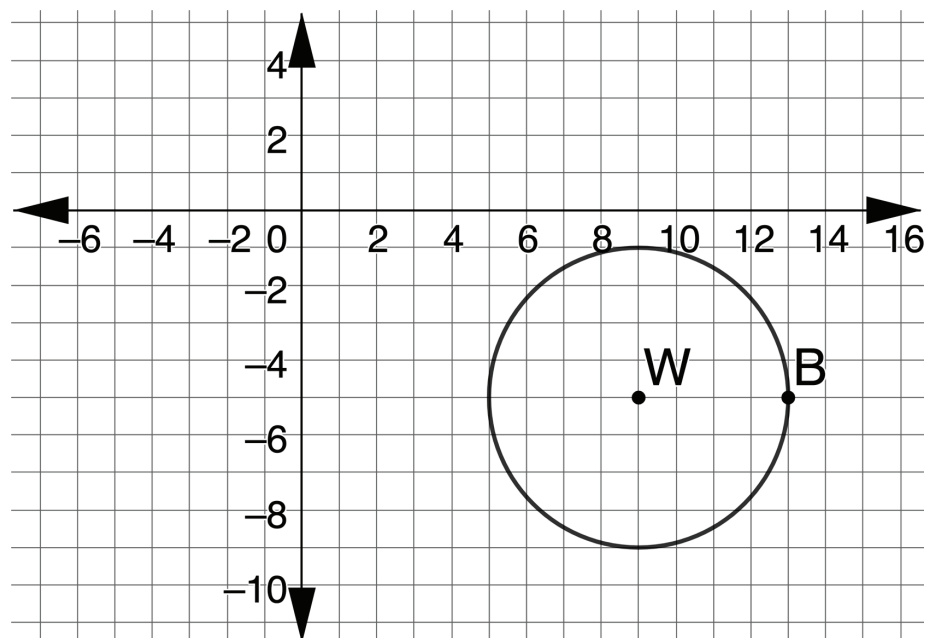




12.7.4 Wrap-Up: Ticket Out the Door

1. Circle W is shown, where point B is on the circle.



- a. Determine the key features of circle W .

Key Features	
Radius	
Center	

- b. Write the equation that represents circle W .

Unit 12, Lesson 8: Standard Form of a Circle



Benchmarks

MA.912.GR.7.2 Given a mathematical or real-world context, derive and create the equation of a circle using key features.

MA.912.GR.7.3 Graph and solve mathematical and real-world problems that are modeled with an equation of a circle. Determine and interpret key features in terms of context.



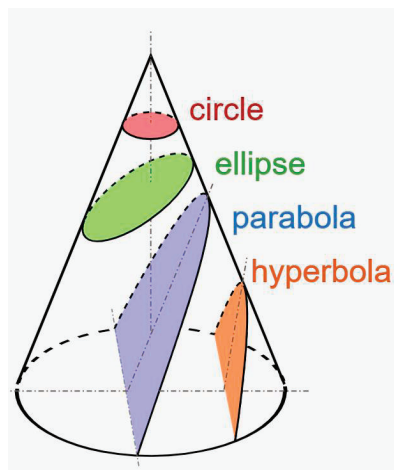
Learning Targets

- ☐ I can rewrite the equation of a circle from standard form to center-radius form.
- ☐ I can determine key features of a circle given the equation in standard form.
- ☐ I can write the domain and range using inequality notation, interval notation, or set-builder notation.



12.8.1 Warm-Up: Conic Sections

Conic sections are the family of curves created when a plane slices through a cone. A circle is best known as a conic section. A diagram of five conic sections is shown.



According to Greek philosopher Proclus, Menaechmus, a member of Plato's Academy, discovered conic sections around 350 B.C. Approximately 100 years later, Apollonius, a Greek geometer and astronomer, developed and published the theory of conic sections. His work earned him the title of the Great Geometer.

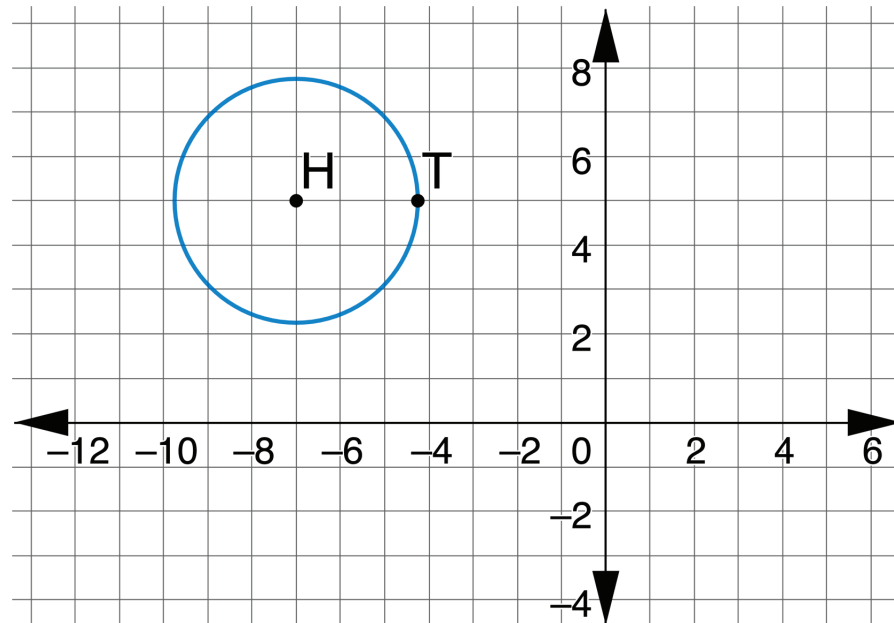


1. Use the diagram to explain why when a cone is sliced horizontally the result is a circle.



12.8.2 Collaboration: Standard Form

1. The center of circle H is located at $(-7, 5)$. Point T lies on circle H at the point $(-4.25, 5)$.



- a. Write the equation of the circle in center-radius form, $(x - h)^2 + (y - k)^2 = r^2$.

- b. Expand each term of the center-radius form to write an equivalent expression.

Term	Equivalent Expression
$(x + 7)^2$	
$(y - 5)^2$	
$(2.75)^2$	



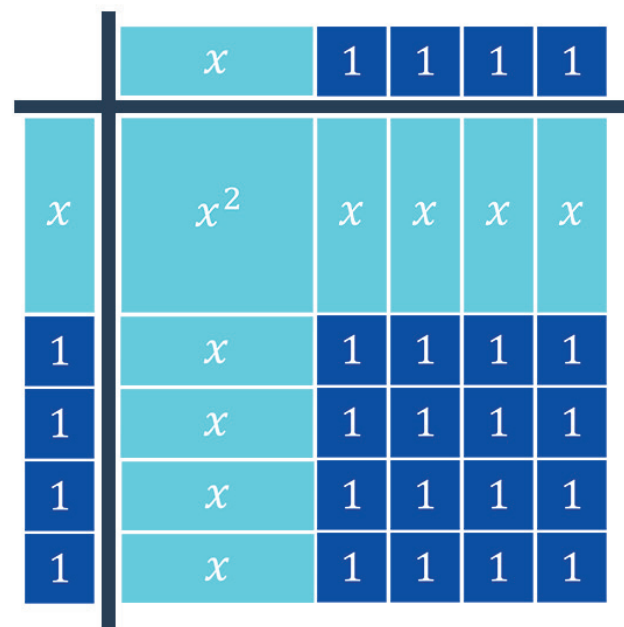
The **standard form** of an equation of a circle is $ax^2 + by^2 + cx + dy + e = 0$.

- c. Use the equivalent expressions of the center-radius form to rewrite the center-radius form of circle H in standard form.



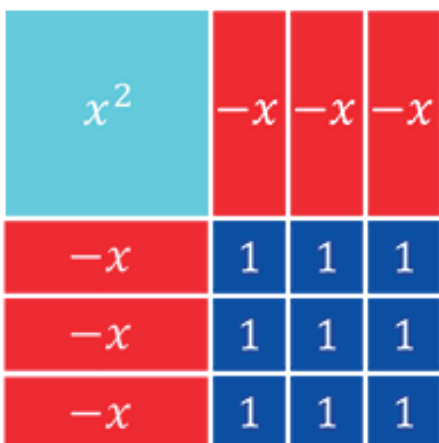
12.8.3 Refresher: Completing the Square

The binomial $(x + 4)^2$ can be rewritten as $x^2 + 8x + 16$, which is a perfect square trinomial. The algebra tiles represent both $(x + 4)^2$ and $x^2 + 8x + 16$.

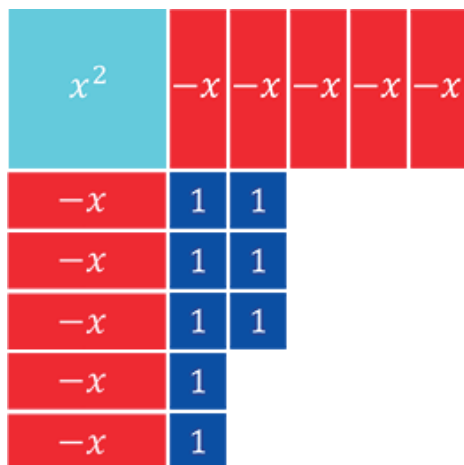


1. Discuss with your partner how the diagram shows that $x^2 + 8x + 16$ is a perfect square trinomial. Summarize your discussion.
2. Discuss with your partner how the diagram shows that $x^2 + 8x + 16 = (x + 4)^2$.

3. Write the perfect square trinomial and the binomial squared that are represented by the diagram.



4. The trinomial $x^2 - 10x + 8$ is shown.



- a. How many 1 tiles have to be added to complete the square?

Completing the square by adding 1 tiles will change the value of the original trinomial unless the same number of -1 tiles are also added.

- b. Complete the table.

Completing the Square	
Original Expression	$x^2 - 10x + 8$
Completing the square	$x^2 - 10x + 8 + \underline{\hspace{1cm}}$  number of 1 tiles added
Adding the opposite so the expression is equivalent to the original expression	$x^2 - 10x + 8 + \underline{\hspace{1cm}} + \underline{\hspace{1cm}}$  number of -1 tiles added
Rewriting so that there is a perfect square trinomial	$(x^2 - 10x + 25) - 17$  perfect square trinomial
Writing the perfect square trinomial as a binomial squared	$(x - 5)^2 - 17$



12.8.4 Guided Instruction: Rewriting the Standard Form of the Equation of a Circle

There are several ways to mathematically represent the domain and range.

Written Description	Inequalities	Set-Builder Notation	Interval Notation
All real numbers greater than or equal to -5 and less than or equal to 5	$-5 \leq x \leq 5$	$\{x: -5 \leq x \leq 5\},$ $\{x -5 \leq x \leq 5\},$ $\{x: x \geq -5 \text{ and } x \leq 5\},$ or $\{x x \geq -5 \text{ and } x \leq 5\}$	$[-5, 5]$

Rewrite each standard form equation of a circle to center-radius form of a circle. Then, give the key features of the circle. Write the domain and range using interval notation and set-builder notation.

1. $x^2 + y^2 - 14x + 6y + 26 = 0$

Center-Radius Form	Center	Radius
	Domain	Range

2. $x^2 + y^2 - 18x - 10y = -52$

Center-Radius Form	Center	Diameter
	Domain	Range



12.8.5 Guided Practice: Converting Forms of a Circle

1. Consider the equation of the circle represented by $(x + 8)^2 + (y + 11)^2 = 14^2$.
 - a. Rewrite $(x + 8)^2 + (y + 11)^2 = 14^2$ in standard form.

- b. Give the key features of $(x + 8)^2 + (y + 11)^2 = 14^2$

Center	Diameter	Radius
Domain	Range	

2. Consider the equation of the circle represented by $x^2 + y^2 - 20x = -36$.
 - a. Rewrite $x^2 + y^2 - 20x = -36$ in center-radius form.

- b. Give the key features of $x^2 + y^2 - 20x = -36$

Center	Diameter	Radius
Domain	Range	



12.8.6 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept.



I need help.



I got it.



I could help someone.

Rewriting center-radius form to standard form			
Key features of a circle			
Completing the square			
Rewriting standard form to center-radius form			
Set-builder notation			
Interval notation			

Unit 12, Lesson 9: Graphing Circles



Benchmark

MA.912.GR.7.3 Graph and solve mathematical and real-world problems that are modeled with an equation of a circle. Determine and interpret key features in terms of context.



Learning Targets

- ☐ I can graph a circle and determine its key features when given an equation of a circle.
- ☐ I can solve mathematical and real-world problems that are modeled with an equation of a circle.